Analysis of Poverty Levels in Eastern Indonesia in 2021 Using the Generalized Least Squares (GLS) Approach to Address Autocorrelation

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***Abstract*—Despite various development initiatives, the welfare gap between Western and Eastern Indonesia remains a critical issue. This study investigates the determinants of poverty across 241 districts in Eastern Indonesia using 2021 data. Initially, the Ordinary Least Squares (OLS) method was applied, but diagnostic tests detected significant autocorrelation, a common issue in spatial data that biases standard error estimates. To correct this and ensure valid inference, the analysis transitioned to a Generalized Least Squares (GLS) model. The results challenge the conventional view that economic growth alone drives poverty reduction; specifically, Regional GDP and open unemployment rates were found to be statistically insignificant in this region. Instead, the findings demonstrate that poverty levels are most responsive to improvements in human capital namely life expectancy and years of schooling and per capita purchasing power. Furthermore, the data reveals a strong positive correlation between the Gini ratio and poverty, suggesting that high income inequality undermines development efforts. These outcomes imply that policy in Eastern Indonesia should pivot from purely seeking economic expansion to prioritizing equitable distribution and the quality of basic services.**

***Keywords*—poverty determinants, eastern indonesia, generalized least squares (GLS), human capital, income inequality.**

1. INTRODUCTION

A. Background

The welfare disparity between Western Indonesia (KBI) and Eastern Indonesia (KTI) remains a persistent development issue. According to 2021 data from the Badan Pusat Statistik (BPS), provinces in Nusa Tenggara, Maluku, and Papua consistently record poverty rates far exceeding the national average [1]. This indicates that the equitable development target outlined in the 2020-2024 National Medium-Term Development Plan (RPJMN) still faces significant structural challenges [2].

A previous study in East Nusa Tenggara by Nalle, Seran, and Bria (2022) found that poverty in the eastern region has unique characteristics that differ from those in Java, where location and accessibility play crucial roles [3]. This characteristic of data between neighboring districts often creates problems with spatial autocorrelation, which, if ignored in standard regression analysis (Osrdinary Least Squares or OLS), can lead to biased conclusions. Gujarati and Porter (2017) assert that violation of this assumption invalidates statistical significance tests [4].

Therefore, a more robust econometric approach is needed. Baltagi (2021) suggests using the Generalized Least Squares (GLS) method to handle data structures with inter-observation correlation and non-constant variance [5]. This study aims to apply the GLS method to precisely analyze poverty in Eastern Indonesia.

B. Problem Foundations

a. How do macroeconomic variables, human capital, and inequality influence poverty levels in Eastern Indonesia in 2021?

b. Is the OLS estimation model biased due to violation of classical assumptions?

c. Is the GLS method proven to be more effective than OLS in modeling poverty levels in regions with high disparities such as KTI?

C. Objectives

a. Analyze the significance of the influence of macroeconomic variables, human capital, and inequality on poverty levels in Eastern Indonesia in 2021.

b. Analyze the OLS estimation method to meet classical assumptions.

c. Analyze the comparative effectiveness of the GLS and OLS method in modeling poverty levels in regions with high inequality such as KTI.

D. Benefits

This research is expected to provide concrete evidence regarding statistically valid determinants of poverty, so that it can serve as a reference for local governments in prioritizing development budgets, both in the physical (economic) and non-physical (health and education) sectors.

2. THEORETICAL STUDY

A. Determinants of Poverty

Multidimensional Perspective: Haughton and Khandker (2016) define poverty as a deprivation of well-being that results in an individual's inability to meet a minimum standard of living [6].

a. Economic Growth (PRDP): Classical development theory states that economic growth (PRDP) will reduce poverty. However, Todaro and Smith (2015) emphasize that growth without equity will only widen inequality [7].

b. Human Capital (Health & Education): Based on Human Capital theory, education and health are vital investments. Mankiw (2019) states that increases in Life Expectancy and Years of Schooling directly increase labor productivity [8].

c. Inequality (Gini Ratio): Recent research by Maskur et al. (2023) shows that, at the same income level, regions with a more unequal income distribution (high Gini Ratio) tend to have higher levels of poverty [9].

d. Employment (Unemployment): Yacoub (2012) in his study found that the Open Unemployment Rate (TPT) is positively correlated with poverty due to the loss of the household's main source of income [10].

B. Empirical Review

Previous empirical studies have yielded mixed results. Pratama (2019) found that macroeconomic variables were highly dominant nationally [11], while Zuhdiyaty and Kaluge (2017) found that social variables were more influential in the short term [12]. Specifically for Eastern Indonesia, Wulandari and Budiyanti (2021) emphasized the importance of human resource quality over physical infrastructure [13]. This aligns with the findings of Puspita and Yanuarti (2020) in Papua, which showed that the determinants of poverty are highly locally specific [14].

3. METHODOLOGY

A. Data and Data Sources

Our team accessed this dataset for the first time in Kaggle on the 26th of November 2025, titled "***Socio-Economic of Indonesia in 2021***" and based on our findings, this dataset was used in the context of machine learning (K Means / Clustering). In the process of making this paper, our team has not found any other research using this dataset in the context of using Regression Analysis to find which predictor variables are significant and build a prediction model. This part was made as a reminder that our team's work is **NOT plagiarized** in any way, shape, or form from other papers that are only used as references and that our findings within this paper are entirely our own.

B. Research Variable

a. Response Variable (Y): poorpeople_percentage (Percentage of poor people).

b. Predictor Variables (X):

i. X_1: reg_gdp (Gross Regional Domestic Product in billions of Rupiah) as an Economic Indicator.

ii. X_2: life_exp (Life expectancy in years) as a Health Indicator.

iii. X_3: avg_schooltime (Average years of schooling in years) as an Education Indicator.

iv. X_4: exp_percap (Per capita expenditure in thousands of Rupiah) as an Indicator of Purchasing Power.

v. X_5: tingkat_pengangguran_terbuka as an Employment Indicator.

vi. X_6: rasio_gini as an Inequality Indicator.

C. Research Variable

The data analysis process was carried out using the RStudio application with the following steps:

a. Initial Modeling: Ordinary Least Squares (OLS)

The first step is to estimate a multiple linear regression model using the OLS method. This method works by minimizing the Sum of Squared Residuals between the actual and predicted values [16]. The resulting model equation is:

$$\boldsymbol{Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i} + \beta_6 X_{6i} + \epsilon_i}$$

Where β_0 is a constant, β_{1-6} are the regression coefficients, and ϵ_i is the error term. To obtain an efficient model, variable selection is carried out using the Backward Elimination method, where variables that are not statistically significant will be removed from the model iteratively until the best model is obtained, as suggested by Chatterjee and Hadi (2013) [17].

b. Classical Assumption Diagnostic Test

Referring to the Gauss-Markov Theorem, the OLS estimator will only be the Best Linear Unbiased Estimator (BLUE), namely an estimator that is unbiased and has minimum variance if all classical assumptions are met [16]. The validity of hypothesis testing (t-test and F-test) is highly dependent on the fulfillment of the following assumptions:

i. Normality: Using the Liliefors (Kolmogorov-Smirnov) test to check the distribution of residuals.

This assumption requires that the residuals (ϵ) must be normally distributed, otherwise there is a violation of the assumption which causes the statistical test to be invalid [16].

ii. Non-Multicollinearity: Using the Variance Inflation Factor (VIF) value to ensure there is no strong correlation between independent variables. This assumption requires that there should be no significant linear relationship between independent variables [16].

iii. Homoscedasticity: Using the Breusch-Pagan test to detect whether the error variance is constant (Homoscedasticity) or changing (Heteroscedasticity). This assumption requires that the variance of the residual must be constant for each value of the independent variable ($Var(\epsilon_i = \sigma^2)$. If the variance changes as the value of X changes, then heteroscedasticity occurs [18].

iv. Non-Autocorrelation: Using the Durbin-Watson test to detect the presence of correlation between errors, which often occurs in data with regional proximity. This assumption requires that there should be no correlation between the residuals in the i-th observation and the j-th observation [19].

In the context of regional socio-economic data, particularly in Eastern Indonesia, violations of the heteroscedasticity assumption often occur due to significant disparities in economic scale between districts/cities. Although the presence of heteroscedasticity does not bias the OLS estimator, it does render it inefficient. This results in biased standard errors, which can ultimately lead to errors in statistical inference.

c. Generalized Least Squares (GLS)

If a violation of classical assumptions (especially heteroscedasticity and autocorrelation) is found in the OLS model, the analysis is continued using the Generalized Least Squares (GLS) method. Unlike OLS which assumes the error variance-covariance matrix as a constant scalar ($\sigma^2 I$), GLS takes into account a more complex error structure. This method performs a linear transformation on the original variables using a weighting matrix (Ω^{-1}) to eliminate inter-observation correlation and stabilize variance [20]. The GLS estimator (β_{GLS}) is calculated using the formula:

$$\boldsymbol{\beta}_{\mathbf{GLS}} = (\boldsymbol{X'\Omega^{-1}X})^{-1}\boldsymbol{X'\Omega^{-1}Y}$$

The use of GLS aims to obtain an efficient and statistically valid estimator, so that hypothesis testing (t-test and F-test) can be trusted even though the data has diverse spatial characteristics [5]. Evaluation of the best model is based on the significance of the variables, the coefficient of determination (R^2), and the information criterion (AIC/BIC).

4. RESULT AND DISCUSSION

A. Variable Analysis

- Correlation Between Predictor Variables

	X1	X2	X3	X4	X5	X6
X1	1	0.03	0.49	0.59	0.2	-0.3
X2	0.03	1	-0.02	-0.03	0.22	0.06
X3	0.49	-0.02	1	0.67	0.28	-0.24
X4	0.59	-0.03	0.67	1	0.19	-0.19
X5	0.2	0.22	0.28	0.19	1	-0.18
X6	-0.3	0.06	-0.24	-0.19	-0.18	1

The results of the correlation analysis between each predictor variable based on the table above shows that each of them does not have a perfect linear relationship with the others, which confirms that no multicollinearity is present within the data. While there is a moderate positive linear relationship between the average school time (avg_schooltime) and expenditure per capita (exp_percap) that is indicated by the pearson coefficient of 0.67, all the variables are still distinct enough to be treated as independent.

- Correlation Between Each Predictor Variables and the Response Variable

	poorpeople_percentage (y)
life_exp (x1)	-0.633
reg_gdp (x2)	0.118
avg_schooltime (x3)	-0.581
exp_percap (x4)	-0.696
tingkat_pengangguran_terbuka (x5)	-0.192
rasio_gini (x6)	0.554

The correlation analysis on the table above shows that directly improving people's everyday lives, especially their spending power (-0.696), health (-0.633), and education (-0.581) is the most effective way to decrease poverty and possibly eliminate them based on the given data. This indicates that poverty rates dramatically decrease when these three variables are improved.

However, the Gini Ratio (0.554) reveals that the percentage of poor people increases as inequality worsens, showing that the distribution of wealth is equally as important as the amount of wealth itself. On the other hand, both the Open Unemployment Rate (tingkat_pengganguran_terbuka) and the Regional GDP (reg_gdp) yields a pearson coefficient of only -0.192 and 0.118, which points out that they have an extremely weak linear relationship with the response variable.

But, this does not mean that reg_gdp and tingkat_pengangguran_terbuka are dropped just because they have a weak linear relationship with the response variable. In multiple linear regression, the decision whether or not to drop certain predictor variables lies in what is called variable selection using all possible subsets or statewise through forward selection as well as backward elimination to achieve the best possible model.

B. Model Building

In this section, we will be diving into the main part which is to build the best possible model to predict the percentage of poor people within Eastern Indonesia. A regression model is said to be perfect if it doesn't contain multicollinearity (VIF < 10) and meets all the error assumptions that is previously explained in the Theoretical Study section.

- Initial Model

	Estimate	**Standard Error**	**t value**	**Pr (>|t|)**
Intercept	0.3257	9.707	3.355	0.000926
X1	0.01047	0.004583	2.284	0.023290
X2	-0.6221	0.1245	-4.996	0.0000115
X3	-0.15317	0.2631	-2.021	0.044442
X4	-0.001465	0.0001799	-8.144	2.28e-14
X5	0.08613	0.2972	0.290	0.772218
X6	114.1	11.34	10.065	< 2e-16

Residual Standard Error		5.128	
Multiple R-Squared	0.7113	**Adjusted R-Squared**	0.7039
F-Statistic	96.07	**p-value**	< 2.2e-16

The table above shows the full summary of the initial model by regressing poorpeople_percentage(y) with all the predictor variables. From the obtained results, we can conclude that all the predictor variables bar the open unemployment rate (tingkat_pengangguran_terbuka) is statistically significant in affecting the percentage of poor people within Eastern Indonesia.

As for the open unemployment rate, since within the initial model is not statistically significant given by the p-value (0.772218 > 0.05) then it is valid to be removed by process of elimination using statewise through backward elimination. Backward elimination works iteratively because it involves doing the same process repeatedly until the ideal model in relation to the dataset used is reached. It starts by regressing the response variable with all the predictor variables and then identifies the least significant predictor variable (the one with the highest p-value and above the pre-determined threshold). After that, it removes that insignificant variable and then re-evaluates itself by fitting a brand new model, now with only the remaining predictor variables. This process will stop if and only if all the remaining predictor variables are deemed significant.

- Partial Model (using Backward Elimination)

	Estimate	**Standard Error**	**t value**	**Pr (>|t|)**
Intercept	32.91	9.617	3.422	0.000734
reg_gdp	0.01078	0.004443	2.427	0.015990
life_exp	-0.6209	0.1242	-4.999	1.13e-06
avg_school time	0.5174	0.2579	-2.006	0.046017
exp_percap	0.001465	0.0001795	-8.163	2.00e-14
rasio_gini	113.7	11.22	10.131	< 2e-16

Residual Standard Error		5.118	
Multiple R-Squared	0.7112	**Adjusted R-Squared**	0.705
F-Statistic	115.07	**p-value**	< 2.2e-16

	X1	**X2**	**X3**	**X4**	**X6**
VIF	1.009569	1.659668	1.878803	2.176327	1.120611

The table above shows the full summary of the partial model after going to the process of backward elimination. From the obtained results, we can conclude that all the remaining predictor variables are statistically significant in affecting the percentage of poor people within Eastern Indonesia. This partial model is also better compared to the initial model since the difference in the value of the R^2 is very small yet requires way less data to gather since one of the predictor variables was dropped and no multicollinearity is present.

Assumptions Test	**p-value**
Liliefors (Kolmogorov-Smirnov) Normality Test	0.02102
Studentized Breusch-Pagan Test	0.0006112
Durbin-Watson test	8.087e-05

The table above are the results of each assumption test after constructing the partial model through the means of backward elimination. The obtained p-value of each test is lower than the established significance level (0.05), which means said model using OLS is not ideal yet in predicting the percentage of poor people within Eastern Indonesia.

- Generalized Least Square Model

After a thorough examination of all the error assumptions, the Generalized Least Squares (GLS) model was chosen, giving a trustworthy and accurate statistical tests precedence over a simple model. On the other hand, the problem of non-normality was extremely persistent and resistant to resolution using methods such as robust and quantile regression.

Although Generalized Least Square (GLS) was used to correct autocorrelation and Weighted Least Squares (WLS) was used to resolve heteroscedasticity, both methods had the unfavorable effect of making one of the predictor variables statistically insignificant, meaning they have to be dropped as well (reg_gdp for GLS and avg_schooltime for WLS). The most serious inferential threat, autocorrelation, which directly invalidates the standard errors and p-values, was successfully addressed by the GLS model, which is why it was selected. Additionally, the GLS solution produced a higher R^2 value, guaranteeing the best explanatory fit while maintaining the relevance of all the significant predictor variables.

	Estimate	**Std. Error**	**t value**	**Pr(>|t|)**
Intercept	20.85595	10.842485	1.923540	0.0556
X1	-0.47038	0.131451	-3.578405	0.0004
X3	-0.69090	0.291086	-2.373519	0.0184
X4	-0.00119	0.000195	-6.089660	0
X6	114.38673	15.497895	7.303978	0

Assumptions Test	**p-value**
Liliefors (Kolmogorov-Smirnov) Normality Test	7.4e-05
Studentized Breusch-Pagan Test	1.067e-06
Durbin-Watson test	0.8509

	X1	**X3**	**X4**	**X6**
VIF	1.467742	2.064256	2.104386	1.049196

The resulting GLS model is both simple and statistically sound. Although the final intercept of the GLS model's became statistically insignificant, this result was anticipated because within the parameters of this investigation, the combination of zero values for all predictor variables is not a theoretically significant possibility. The validity and interpretation of the slope

coefficients for the other significant predictors are unaffected by this, though.

Residual Standard Error		5.225091	
Multiple R-Squared		0.707800673403728	
AIC	1475.542	**BIC**	1503.218

The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values of the GLS model were higher than those of the WLS model, but this trade-off is considered acceptable. In order to guarantee that the GLS corrected estimates are the Best Linear Unbiased Estimators (BLUE), the validity of the hypothesis testing and the preservation of all theoretically required variables were given the most attention.

5. CONCLUSION

This study establishes that standard regression models are insufficient for analyzing poverty in Eastern Indonesia due to the presence of spatial autocorrelation, which violates classical OLS assumptions. By adopting the Generalized Least Squares (GLS) method, we provide a more robust statistical basis for understanding regional welfare dynamics.

The empirical evidence leads to two primary conclusions. First, the macroeconomic scale represented by Regional GDP does not automatically translate into poverty reduction in Eastern Indonesia. The statistical insignificance of GDP suggests that economic growth in these districts has not been inclusive enough to reach the lowest income deciles. Second, the fight against poverty is heavily dependent on "non-income" factors. Life expectancy, education duration, and purchasing power are proven to be the most effective levers for lowering poverty rates, whereas widening inequality (Gini Ratio) significantly hinders these efforts. Consequently, local governments should prioritize budget allocations for health and education infrastructure over generic economic stimulus, ensuring that growth is accompanied by equity.

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